

PANDAS (X)

numpy is used for working with arrays. pandas is used for working with structured data like tables. before training every AI model we first clean and analyse the data using pandas.

pandas → open source python library used for data manipulation, data analysis, and data cleaning helps us to work with structured data in the form of rows and columns.

pandas → used to read, organize, clean, analyze and manipulate datasets.

Data cleaning:

- Remove null values

- Remove Duplicates

- Replace incorrect values

- Handle missing data.

Data Analysis:

pandas provide Function For

- Average

- Maximum

- Minimum

- Sum

- Grouping

- Sorting

- Filtering

Data visualization Support:

- matplotlib

- Seaborn

installing pandas:

pip install pandas
importing pandas:
import panda as pd

2 Main datastructure:

Series

Dataframe

Series:

⇒ A one dimensional data structure.

Dataframe:

⇒ Two dimensional table

Series:

A one dimensional labeled array capable of storing any type of data (one column of data)

import pandas as pd

Marks = pd.Series([80, 90, 75, 95])

print (marks)

output:

0 80

1 90

2 75

3 95

dtype: int64

Series with custom index:

marks = pd.Series([80, 90, 75], index=["John", "David", "Sara"])

print (marks)

output:

John 80

David 90

Sara 75

dtype: int64

Accessing Values From Series:

print(marks["John"])

output:

80

Dataframe:

It's a two dimensional labeled data structure consisting of rows and columns.

Creating a dataframe (df)

```
import pandas as pd
data = {
    "name": ["John", "David", "Sara"],
    "age": [25, 30, 28],
    "city": ["chennai", "mumbai", "delhi"]}
df = pd.DataFrame(data)
print(df)
```

Output:

	name	age	city
0	John	25	chennai
1	David	30	Mumbai
2	Sara	28	Delhi

Qns:

1. Create a series containing 10 20 30 40 50
2. Create a series with custom index
apple banana orange
3. Create the following dataframe?
(in table = column 4 rows)
name, age, ~~category~~ city
aasha, Ragul, Priya, 21, 24, 23, Chennai, bangalore
Hyderabad.

CSV File:

- * Comma Separated Values
- * Each row Represents One record
- * Each Column Represents one Feature

Reading a CSV File:

`pd.read_csv()`

Syntax:

```
import pandas as pd
df = pd.read_csv("students.csv")
print(df)
```

head() (first)

head displays the first five rows of dataset

```
print(df.head())
```

head(n)

`print(df.head(3))` → first 3 rows

Tail() (last)

Displays the last five rows

```
print(df.tail())
```

Tail(n)

```
print(df.tail(2))
```

shape

Returns the number of rows and Columns.

```
print(df.shape)
```

output:
(100, 4)

ndim:

Returns the number of dimensions.

`print(df.ndim)`

Datframe is always two dimensional.

Size:

Returns the total number of elements

`print(df.size)`

columns:

Returns all column names

`print(df.columns)`

Index:

Returns Row indexes

`print(df.index)`

dtype:

Returns the datatype of each column

`print(df.dtypes)`

name	object	→ string condition as
age	int64	
city	object	
salary	int64	

INFO() (missing values display)

One of the most important functions

`df.info`

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 100 entries

columns: data columns

data	columns
name	object
age	int64
city	object
salary	int64

[tells us :
no of rows
no of column
Datatype
missing values
memory usage]

Describe()

Returns the statistical Summary
print (df.describe())

output:

Age salary

count → number of values
Mean → Avg
std → Standard How much data spread
min → smallest
25% → first quartile (25% of data lies below this value)
50% → median value
75% → 3rd quartile (75% of data lies below this value)

Sample()

Returns Random rows

print (df.sample(5)) → random 5 rows coming

value_counts()

(use to customer, analysis, machine learn)

Count Repeated values

print (df["city"].value_counts())

output:

chennai 3

mumbai 2

delhi 1

Unique()

Returns Unique value

print (df["city"].unique())

output:

['chennai', 'mumbai', 'delhi']

N Unique()

Returns the number of unique values

print (df ["city"], unique())
output:
3

Memory Usage:

print (df. memory_usage())

* large dataset handling - tgm

Customer Review Dataset:



read the csv file

head

info → check missing columns

describe → analyse numerical columns

value_count - count positive and negative Reviews

proceed to processing and model training.

Qn:

1. load a csv file called students.csv
2. Display first 5 rows and last 5 rows
3. Find no of rows and no of columns
4. Display all column names
5. Find the datatype of every columns
6. Generate the statistical summary
7. Display three Random rows
8. Find how many students belong to each city using value_counts: